

Iterative Model Development for Skin Cancer Malignancy Classification

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Abstract

This project aims to develop a high-accuracy classification model for distinguishing between malignant and benign tumors. This necessitates an iterative development process, evaluating various model families and optimizing hyperparameters to maximize predictive performance. The dataset, sourced from Kaggle, contains synthetically generated demographic, lifestyle, and physiological features. Our top-performing model was Logistic Regression model with Lasso (L1) regularization, utilizing a rigorous feature selection scheme based on chi-squared and t-tests. This model achieved a final testing accuracy of 60.650%, which tied our team for 10th on the Kaggle leaderboard.

1 Introduction

It is estimated that one-in-five Americans will be diagnosed with skin cancer by age 70, which makes it the most common type of cancer in the United States. Further, the World Health Organization estimates that between 115,000 and 120,000 individuals die from skin cancer annually. Importantly, there is a sharp distinction between non-melanoma skin cancers and melanoma. Non-melanoma skin cancers are very common but usually less aggressive, whereas melanoma accounts for only around 1% of skin cancer cases. Yet, these cases of melanoma account for the vast majority of fatalities related to skin cancer.

This disparity highlights the essential role of early detection in skin cancer treatment. Data from the American Skin Cancer Foundation shows that in these cases, the five-year survival rate for localized melanoma is around 99%. However, if these cases are not detected early and become more advanced spreading to distant organs, the five-year survival rate plummets to approximately 35%. Given this data, it is evident that distinguishing between malignant and benign cases can guide treatment and save lives. These high stakes, in tandem with a wealth of biomedical data, have led to a rise in the prevalence of machine learning and data-driven approaches for determining individual patients' malignancy and for supporting researchers in understanding the underlying biological mechanisms of skin cancer.

The objective of this project is to develop the most accurate model possible in order to distinguish between benign and malignant cases. This will include evaluating a range of classification models, with a focus on balancing accuracy and interpretability. Although we aim for these models to be as accurate as possible, it is also important that we understand their results in context so they can provide broader guidance to medical practitioners on the factors driving these predictions. The rest of this paper will cover an initial overview of the data, data pre-processing, iterative model development, a discussion of our results, and some limitations imposed by the data.

2 Exploratory Data Analysis

To begin the modeling process, exploratory data analysis (EDA) is essential for developing an understanding of the dataset. EDA helps reveal patterns and potential relationships among variables, providing insight that guides later cleaning, feature selection, and model development.

We began our EDA by examining 20 numerical variables, including age, income, and daily UV exposure. Histograms identified distribution shapes, outliers, and variables that appeared to be categorical, like `outdoor_job`. We also used density plots to compare numerical variables, such as age, between benign and malignant groups to assess predictive value.

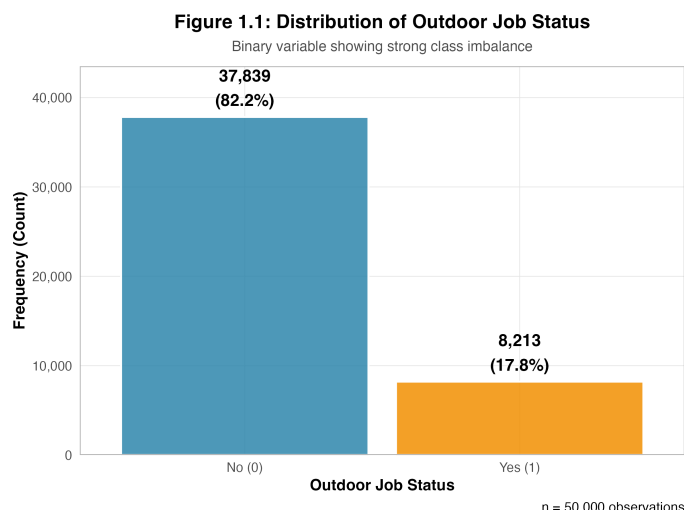


Figure 1: Distribution of Outdoor Job Status

Here we see `outdoor_job` demonstrating a definitive skew towards "No." This indicates that we may have to deal with some feature imbalance issues for the `outdoor_job` predictor and other predictors associated with it. Furthermore, in the age density plot, malignant cases show a slight shift towards older ages

(Figure 1.2). This indicates that age possibly contributes to predictive ability, though the amount of overlap can cause noise in this predictive ability and may weaken it when used by itself.

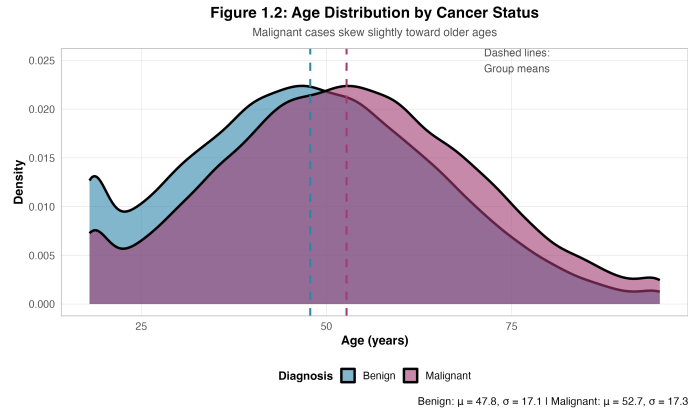


Figure 2: Age Distribution By Cancer Status

Next, we moved to categorical EDA, which focused on understanding both the distribution of categories within each variable and how those categories may relate to skin cancer malignancy. First, the overall frequency of each category was analyzed to identify feature imbalances, outliers, or even improperly encoded categories. For example, `skin_tone` shows variation in how frequently each tone appears, with Fair, Olive, and Medium being the most common categories (Figure 1.3). Understanding these distributions helps clarify which categories are well-represented and which may contribute noise to the model.

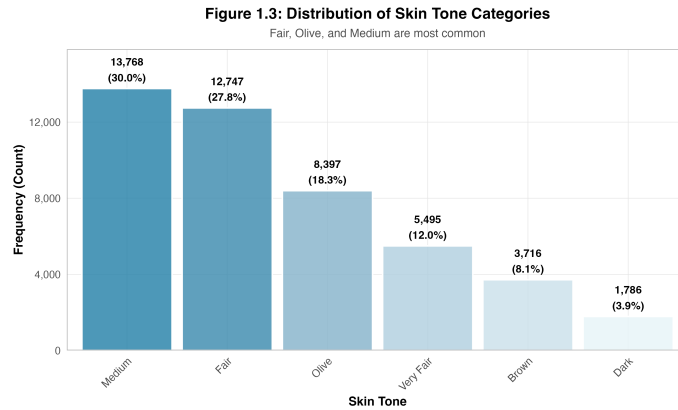


Figure 3: Distribution of Skin Tone Categories

We then evaluated how cancer rates differed across categories using proportion plots. The `hat_use` variable illustrates this relationship by showing

the proportion of benign and malignant cases within each usage level (Figure 1.4). While the use of a hat does not demonstrate strong separation between outcomes, the slight fluctuations across categories suggest potential weak associations that can be explored further in our model testing.

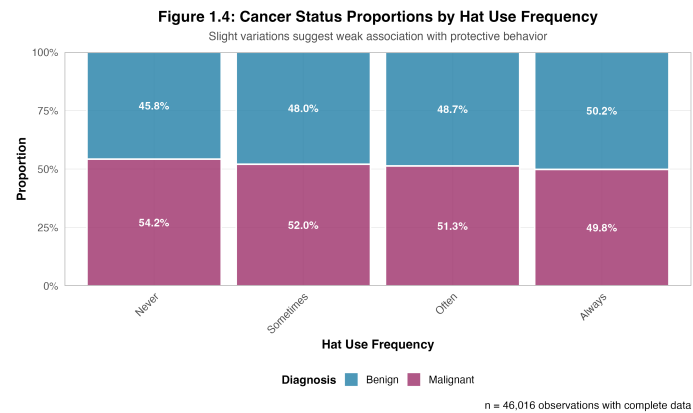


Figure 4: Enter Caption

Finally, we created a series of visuals to investigate the potential of multicollinearity in our data along with the relationships between features. The correlation matrix reveals an exceptionally clean data structure with most predictors showing near-zero relationships ($|r| < 0.10$). The lack of any correlation exceeding 0.30 ensures that each variable contributes unique information to the model. This suggests that when developing our models coefficients will remain stable and the need for dimensionality reduction using techniques like Principal Components Analysis is unnecessary.

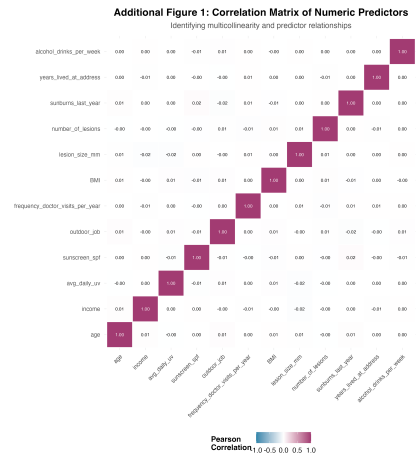


Figure 5: Correlation Matrix of Numeric Predictors

The analysis of Figure 6 reveals a negligible correlation between average daily UV exposure and lesion size for both benign ($r = -0.0018$) and malignant ($r = -0.0203$) diagnoses. Both groups display nearly identical scatter patterns and a wide variance in lesion size regardless of UV intensity. From a clinical perspective, while UV exposure is a known risk factor for cancer development, it does not appear to modulate the physical size of resulting lesions. Consequently, we experimented with interaction terms in the model development process including terms like ($UV \times lesion_size$).

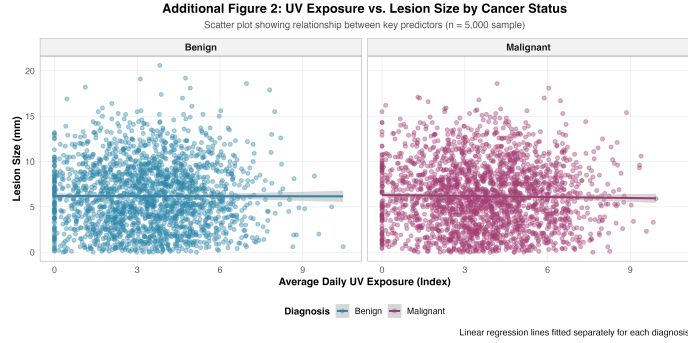


Figure 6: UV Exposure vs Lesion Size By Cancer Status

3 Methods

3.1 Missing Data Imputation

An initial audit of the dataset revealed that all features exhibit missing values with a relatively uniform missingness rate of approximately 8% (ranging from 7.8% to 8.3%). This consistent pattern is exemplified by variables such as *vitamin_d_supplement* (8.3%), *phone_brand* (8.2%), and *skin_tone* (8.2%). To evaluate the nature of this missingness, we conducted preliminary tests for the Missing Completely At Random (MCAR) assumption.

To ensure data integrity, we first characterized missingness patterns using missingness maps, Little’s MCAR test, and proportion tests to compare missingness rates between benign and malignant cases. Our findings revealed a uniform missingness of approximately 8% across most variables, with no significant differences based on outcome ($\chi^2 = 2.14, p = 0.14$).

We evaluated several imputation strategies with varying levels of complexity to address these gaps. Mean and median imputation were considered for numeric variables due to their computational efficiency. For categorical data, mode imputation was utilized to assign the most frequent category. While these methods preserve sample size, they inherently underestimate variance and ignore inter-variable correlations. To capture more complex data structures, we

assessed K-Nearest Neighbors (KNN) imputation, which identifies k neighbors based on the Euclidean distance, as well as Multivariate Imputation by Chained Equations (MICE).

In addition to these methods, we evaluated Random Forest (RF) imputation, specifically utilizing the MissForest algorithm. Unlike MICE, which often relies on linear assumptions, Random Forest imputation is particularly robust for capturing non-linear relationships and complex interactions.

3.2 Variable Selection

To enhance model interpretability, we refined our feature set by removing 20 variables based on three criteria. We first excluded variables that lack a known biological mechanism linking them to oncogenesis, which is the biological process underlying the transformation of healthy cells into cancer cells. Established literature identifies UV exposure, phenotypic characteristics, and family history as the primary drivers of melanoma risk. Variables unrelated to these mechanisms such as *phone_brand*, *music_genre*, and *preferred_shoe_type* were immediately removed to enhance model interpretability and prevent our models from detecting false correlations.

Variables demonstrating poor predictive power in univariate logistic regression ($p > 0.20$), such as *vitamin_d_supplement* ($p = 0.34$) and *near_high_power_cables* ($p = 0.58$), were eliminated. Furthermore, while geographic coordinates including *residence_lat* and *residence_lon* offer potential proxy data for UV exposure, we are unable to work with them unless they are translated using coordinate reference systems and were thus removed.

The impact of this dimensionality reduction was positive across all evaluated algorithms. Reducing the feature count improved the Area Under the Curve (AUC) by up to 1.2% and significantly decreased computational overhead, with training times dropping by 14–31%.

3.3 Feature Encoding

Our dataset comprises three primary categorical variables including nominal, ordinal, and binary. We adopted a unique encoding strategy for each of our models. For Logistic Regression and LDA, ordinal variables were mapped to integers in order to preserve the natural ordering of ordinal data. For nominal variables we created a series of dummy variables with a reference group. Throughout this paper "dummy variables" are synonymous with one-hot encoding.

Tree-based models, such as Random Forest and XGBoost, utilized a hybrid approach. While XGBoost exploited the ordinal structure for monotonic relationships, Random Forest relied on one-hot encoding to avoid split bias toward high-cardinality directions. A specialized ordinal mapping was applied to *family_history*, where "Unknown" was retained as a distinct level (None $\rightarrow$ 0, Unknown $\rightarrow$ 1, 2nd-degree $\rightarrow$ 2, 1st-degree $\rightarrow$ 3). This preserves potential informative missingness, as awareness of family history is often correlated with

the risk itself. Finally, for K-Nearest Neighbors (KNN) all categorical variables were kept as factors.

Table 1: Encoding Strategy by Algorithm and Variable Type

Algorithm	Nominal	Ordinal	Binary	Rationale
Logistic Regression	One-Hot	Ordinal (0-4)	0/1	Preserves linear trends
Random Forest	One-Hot	One-Hot	0/1	Prevents split bias
LDA	One-Hot	Ordinal (0-4)	0/1	Aids normality assumption
XGBoost	One-Hot	Ordinal (0-4)	0/1	Exploits monotonic structure
KNN	One-Hot	Ordinal (0-4)	0/1	Equalizes distance weighting

3.4 Feature Standardization and Scaling

Numeric predictors exhibited vastly different scales, such as *age* (18–95 years), *BMI* (7.4–46.7 kg/m²), and *income* (\$1,343–\$966,000). These disparities can bias distance-based algorithms and affect regularization with scale dependent penalties. To address this, we utilized Z-score normalization.

For distance-based models like KNN and LDA, standardization is critical to prevent high-magnitude variables from dominating distance metrics. For regularized Logistic Regression, it ensures that the L1/L2 penalties are applied uniformly across features. In contrast, tree-based models like Random Forest and XGBoost remain scale-invariant, as their node partition logic is based on ordered feature values, which do not change when standardizing using monotonic functions like Z-scores or $\ln$.

Table 2: Standardization Requirements by Algorithm

Algorithm	Required?	Method	Rationale
Logistic Regression	Helpful	Z-score	Scale-invariant regularization
Random Forest	No	None	Invariant to monotonic transforms
LDA	Yes	Z-score	Prevents variance dominance
KNN	Yes	Z-score	Equalizes distance contributions
XGBoost	No	None	Tree-based scale invariance

3.5 Feature Engineering

To move beyond purely additive relationships, we introduced several interaction and polynomial terms. This effort was motivated by our findings in the literature that factors like family history, lesion size, and lesion color often yield far more predictive power than factors like demographic data. For example, the interaction between immunosuppression and UV exposure ($\text{immuno} \times \text{UV}$) captures the exponentially higher risk faced by patients who suffer from autoimmune disorders. We also calculated proxies for lifetime exposure, such as *cumulative_uv* ($\text{age} \times \text{avg_daily_uv}$).

Furthermore, we introduced polynomial features to account for non-linear dose-response curves. While standard models assume a constant rate of risk increase, quadratic terms like age^2 and lesion_size^2 allow the model to reflect accelerated risk factors observed in the literature. All of these features were applied to algorithms capable of performing automatic feature selection in order to minimize added dimensions.

4 Results

4.1 Logistic Regression

When developing our logistic regression models, we implemented a specific variable selection scheme designed for linear models. Logistic regression assumes that the relationship between the predictors and the log odds of the response is linear, so we ran tests to check for linear associations among our features and to test for differences in means.

For the categorical variables, we applied the Chi-squared test of independence. For each categorical variable, we created the contingency table with the response variable and computed the Chi-squared statistic. Variables with p-values less than 0.05 were considered significantly associated with cancer and retained, while variables with p-values greater than 0.05 were dropped.

Table 3: Chi-Squared Test of Independence for Categorical Predictors

Variable	p-value	Significant
family_history	4.1483×10^{-116}	Yes
skin_tone	1.4635×10^{-79}	Yes
sunscreen_freq	7.7025×10^{-45}	Yes
immunosuppressed	1.2469×10^{-40}	Yes
outdoor_job	1.5175×10^{-24}	Yes
occupation	4.7450×10^{-22}	Yes
tanning_bed_use	1.0325×10^{-17}	Yes
clothing_protection	1.1962×10^{-13}	Yes
skin_photosensitivity	9.1577×10^{-13}	Yes
hat_use	2.1990×10^{-08}	Yes

For the numerical variables, we used the two-sample t-test to compare the mean values between the two target classes (benign and malignant). Each test evaluated if the mean of a numerical predictor differed significantly between the two groups. Variables with p-values less than 0.05 were considered significantly associated with cancer and retained, while variables with p-values greater than 0.05 were dropped. The t-test showed that the *years_lived_at_address* and *income* are significant predictors, but we were doubtful of this result.

We conducted Lasso (L1) regression using all numerical predictors to confirm whether these two predictors are indeed significant. The Lasso regression did not

Table 4: Two-Sample T-Test Results for Numerical Predictors

Variable	p-value	Significant
age	2.2089×10^{-203}	Yes
avg_daily_uv	4.5160×10^{-41}	Yes
number_of_lesions	1.0295×10^{-32}	Yes
sunburns_last_year	4.4276×10^{-27}	Yes
lesion_size_mm	5.8228×10^{-07}	Yes
sunscreen_spf	6.6880×10^{-05}	Yes
years_lived_at_address	0.0247	Yes
income	0.0423	Yes

Table 5: Lasso Regression Coefficient Results for Numerical Predictors

Variable	Coefficient	Selected
(Intercept)	-1.4779	Yes
number_of_lesions	0.1299	Yes
sunburns_last_year	0.1234	Yes
avg_daily_uv	0.0652	Yes
age	0.0164	Yes
lesion_size_mm	0.0114	Yes
exercise_freq_per_week	0.0030	Yes
years_lived_at_address	0.0022	Yes
zip_code_last_digit	0.0016	Yes
sunscreen_spf	-0.0016	Yes
desk_height_cm	0.0013	Yes
residence_lon	-0.0004	Yes
distance_from_beach_km	6.5440×10^{-05}	Yes
income	2.4000×10^{-07}	Yes

set the coefficients of *years_lived_at_address* and *income* to zero, confirming that they are significant predictors. As a result, we decided to keep these predictors in the final model.

Since we have a binary classification problem with far more observations than predictors, we chose Logistic Regression as our baseline model. To prepare the data, all categorical predictors were one-hot encoded. Initial training using Logistic Regression with 10-fold cross-validation resulted in a training accuracy of 0.59668. Furthermore, we conducted diagnostic checks to ensure there were no significant multicollinearity problems within the model. All retained predictors exhibited VIF values well below the common threshold of 5 or 10, with the highest value being 3.6336 for *skin_tone_Medium*, confirming that no significant multicollinearity issues compromised the stability of our baseline model.

Upon submitting the testing set predictions to Kaggle, the model achieved

Table 6: Variance Inflation Factor (VIF) Results for Logistic Regression Model

Variable	VIF
skin_tone_Medium	3.6336
skin_tone_Fair	3.4614
skin_photosensitivity_Low	2.9308
skin_photosensitivity_Moderate	2.9306
skin_tone_Olive	2.7348
occupation_Office	2.5758
hat_use_Never	2.3379
sunscreen_freq_Often	2.2879
skin_tone_Very Fair	2.1829
occupation_Service	2.1463

an accuracy of 0.60650, which was our overall highest-performing result. In an attempt to capture more complex relationships, we introduced several interaction terms, including $avg_daily_uv \times sunscreen_spf$, $hat_use \times outdoor_job$, and $skin_photosensitivity \times sunburns_last_year$. However, this interaction-based approach resulted in a decreased testing accuracy of 0.60430. Subsequent attempts to refine the model—including adding and removing additional interaction terms, experimenting with various regularization schemes with a particular emphasis on Elastic Net, and introducing higher-order polynomial terms—did not surpass the performance of the original Lasso-regularized Logistic Regression model.

4.2 Linear Discriminant Analysis

After seeing our best results consistently come from logistic regression models under a variety of regularization schemes, feature selection techniques, and interaction terms, we began to hit a ceiling in improving accuracy. This motivated us to experiment with linear discriminant analysis models. This was further motivated by the relatively even distribution of features we observed during our EDA that directly follows LDA’s assumption that the data is approximately normally distributed.

Our best LDA model achieved a testing accuracy of 60.0455%, which was slightly worse than that of our best logistic regression model. The diagnostic evaluation of the Linear Discriminant Analysis (LDA) model reveals a mix of met statistical assumptions and significant violations that likely limit its comparative performance. Multivariate normality is partially violated as evidenced by slight deviations in the Q-Q plots. While *age* and *avg_daily_uv* follow the theoretical normal distribution, *lesion_size_mm* exhibits substantial right-skewness, and *number_of_lesions* shows a discrete pattern. Conversely, the assumption of homoscedasticity is largely satisfied. The covariance ellipses for the Benign and Malignant classes show similar orientation and scale and are supported by a log

covariance ratio of -0.58.

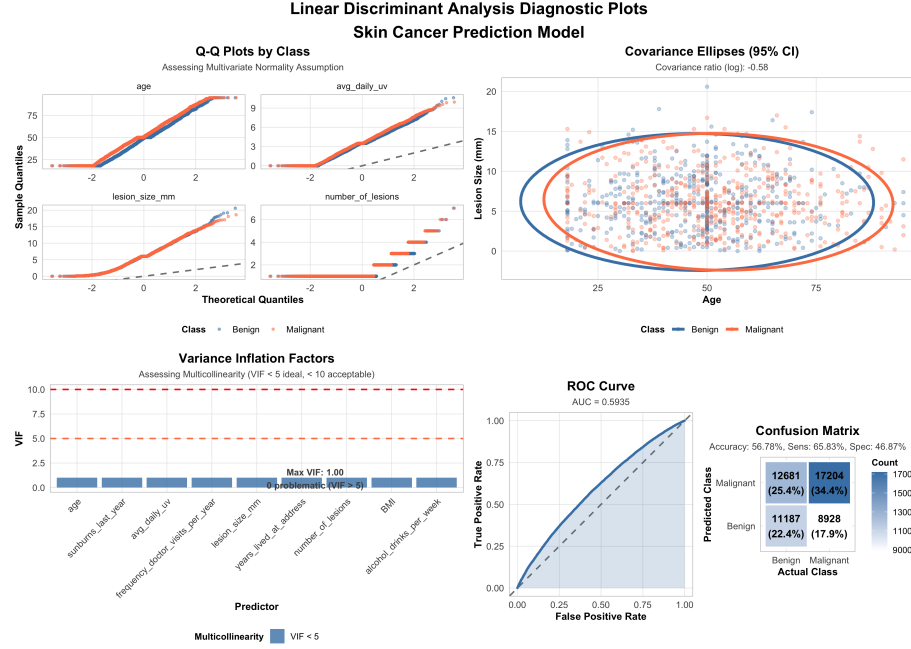


Figure 7: LDA Diagnostics and Cross-Validation Results

4.3 Random Forest

We initially believed that the boundary separating benign and malignant observations would be rather complex. This hypothesis led us to experiment with several ensemble methods, beginning with random forests, to establish a performance benchmark. We experimented with various random forest models that came about from extensive hyperparameter tuning. During the first stage we experimented with feature set sizes ranging from 10 to 50 and evaluated them using 5-fold cross validation to find the optimal subset of features.

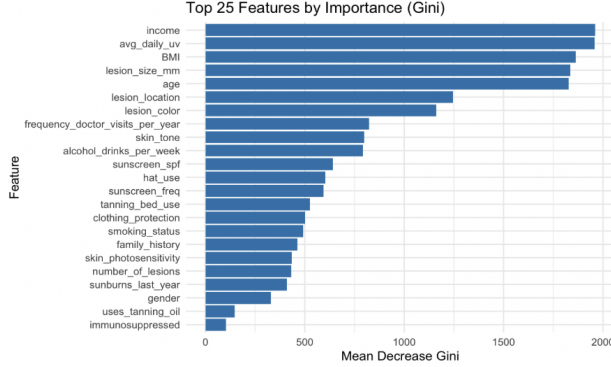


Figure 8: Random Forest Feature Importance Plot

Subsequently, hyperparameters were tuned using a grid search over $mtry$ (calculated based on $\sqrt{p}$, $1.5\sqrt{p}$, and $p/3$), $nodesize$ (1, 5, 10), and $ntree$ (500, 1000, 2000). The evaluation also employed 5-fold stratified cross-validation with fixed folds, where the best configuration was selected based on mean cross-validation accuracy. This approach yielded significant performance gains over baseline models that relied on default $mtry$ ($\sqrt{p}$) and fixed $ntree$ (500) settings. The final model incorporated the optimized hyperparameters trained on the full dataset, with $ntree$ increased to 2000 for the final fit. Our top performing random forest model achieved an accuracy of 59.105%.

4.4 XGBoost

Due to unexpectedly poor performance of our random forest models, we wanted to further explore ensemble methods, particularly Extreme Gradient Boosting (XGBoost). Compared to random forests, XGBoost provides a broader set of regularization options, including built-in L1 and L2 regularization via the `reg_alpha` and `reg_lambda` parameters as well as the `gamma` parameter to ensure node splits occur only when they significantly improve model performance. We used R’s built-in methods to perform a grid search over these parameters, but saw similar performance to the random forest model. Our final XGBoost model achieved 60.110% accuracy and prompted us to sharpen our focus on improving our logistic regression models through further hyperparameter tuning, regularization, and variable selection.

4.5 KNN

Building off of our initial beliefs that the decision boundary would be quite complex we continued to experiment with models that produced non-linear decision boundaries.

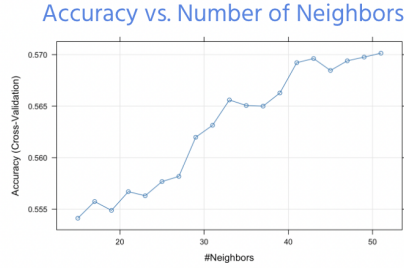


Figure 9: KNN CV-Accuracy Vs Number of Neighbors

The accuracy of the KNN model continued to increase with the number of neighbors until the code’s execution time became prohibitively long. This suggests that the boundary between cancer status is indeed closer to linear, which explains why logistic regression model outperforms more complex models. When comparing these results with the significant improvements observed with various logistic regression models, this suggests that there is significant overlap between benign and malignant cases in the feature space. Furthermore, KNN is a local learner, meaning that its classifications are based on the similarity of a single observation to the k most similar observations. On the other hand, logistic regression is a global learner that analyzes trends across the entire dataset. Our results suggest that KNN struggles at lower values of k due to a lack of global information, as accuracy consistently increases with k . In the end, our best KNN model attained a testing accuracy of 57.480%.

5 Limitations and Future Work

Several limitations constrained the predictive performance of the models developed in this project. First, the heavy reliance on self-reported data introduces potential bias and measurement inaccuracies, which can contribute substantial noise to the dataset. While many of these variables capture behavioral or demographic features associated with skin cancer risk, they do not directly reflect the biological nature that may lead to the development of skin cancer. Additionally, the absence of clinical imaging and medical test results further limits the models’ ability to distinguish between benign and malignant cases.

Early exploratory data analysis also revealed substantial overlap in cancer status across nearly all predictors. This overlap indicates that the available features do not allow for clear separation between benign and malignant, and that the predictive signal is often obscured by noise. As a result, no single predictor, or, in many cases, a combination of predictors, provides strong, standout predictive power for the models. This stands in stark contrast more realistic clinical data that shows an enormous skew towards benign cases. This presents its own challenges with simple linear models like logistic regression often struggling to find patterns in the malignant cases.

Finally, the volume of missing data across multiple variables introduces additional uncertainty. Although several imputation strategies and feature engineering approaches were used, these methods cannot fully recover missing information. As a result, imputed values may not accurately reflect the true underlying data.

For future research, numerous improvements can be made to the models for this project. This includes incorporating clinical information into the data in order to allow our models to capture risks directly linked to skin cancer, rather than relying on demographic and self-reported factors.

Further testing of feature engineering could also improve the model. Higher-order interactions and transformations guided by dermatological research could be explored. These changes, when combined, could allow complex relationships to form.

Finally, future work could explore more computationally intensive modeling approaches, such as extensive hyperparameter tuning. However, given that the data consist mainly of survey-style variables and that the individual predictors are weak, improvement may be minimal.

6 Conclusion

This project set out to build a predictive model for identifying malignant skin cancer using a large dataset of behavioral, demographic, and environmental variables. Through extensive EDA, feature selection using chi-square tests, t-tests, and regularization schemes, logistic regression with cross-validation emerged as the most effective method.

Although more flexible models such as random forests and XGBoost were explored, they did not outperform logistic regression in test accuracy or Kaggle performance. The best model achieved a Kaggle accuracy of 0.60650.

Our findings indicate that the predictors in this dataset have only moderate power to distinguish malignant from benign cases. The modeling process highlights how thorough EDA, feature selection, and model evaluation are crucial for building a reliable final model.

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